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1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out,

"Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get

really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The

villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock

into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

Answer:

For this case study, think of the **boy's warning ("Wolf!")** as the **prediction** and the **actual presence of the wolf** as the **real condition**.

Definitions

- **True Positive (TP):** Wolf is present, and the boy cries "Wolf!"
- **False Positive (FP):** Wolf is not present, but the boy cries "Wolf!"
- **False Negative (FN):** Wolf is present, but no warning is given.
- **True Negative (TN):** Wolf is not present, and no warning is given.

Applying to the Story

Situation	Actual Wolf	Boy Cries "Wolf!"	Classification
The boy cries "Wolf!" when there is no wolf (first few times)	No	Yes	**False Positive (FP)**
The boy cries "Wolf!" when there is a real wolf	Yes	Yes	**True Positive (TP)**
If there was no wolf and the boy did not cry "Wolf!" ***(not mentioned in the story)***	No	No	**True Negative (TN)**

If there was a wolf and the boy did not cry "Wolf!" ***(not mentioned in the story)***	Yes	No	**False Negative (FN)**

Final Answer

- **TP (True Positive):** The real wolf appears, and the boy correctly cries **"Wolf!"**
- **FP (False Positive):** The boy cries **"Wolf!"** when there is **no wolf** (the prank).
- **FN (False Negative):** Not present in this story. It would occur if the wolf came and the boy did **not** warn anyone.
- **TN (True Negative):** Not present in this story. It would occur if there was no wolf and the boy did **not** cry "Wolf!".

Confusion Matrix

	Actual Wolf	No Wolf
Boy cries "Wolf!"	**TP**	**FP**
Boy does not cry	**FN**	**TN**

Result

- **True Positive (TP):** The shepherd boy cries **"Wolf!"** when a real wolf is present.
- **False Positive (FP):** The shepherd boy cries **"Wolf!"** when there is no wolf (false alarm).
- **False Negative (FN):** Not present in the story. It would occur if the wolf came and the boy failed to warn the villagers.
- **True Negative (TN):** Not present in the story. It would occur if there was no wolf and the boy did not cry **"Wolf!"**.

Conclusion:
This case study demonstrates the importance of minimizing **False Positives**.

Repeated false alarms reduce trust, causing genuine warnings to be ignored, which can lead to serious consequences.

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning

model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat

in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall

and F1 Score .

Answer:

Introduction

In Machine Learning, a **Confusion Matrix** is used to evaluate the performance of a classification model. It consists of four outcomes:

- **True Positive (TP):** The model correctly predicts the positive class.
- **False Positive (FP):** The model incorrectly predicts the positive class.
- **False Negative (FN):** The model incorrectly predicts the negative class.
- **True Negative (TN):** The model correctly predicts the negative class.

Step 1: Given Data

- Total Images = **100**
- Cat Images (Actual Positive) = **30**
- No Cat Images (Actual Negative) = **70**

The model correctly predicts cats in **25** images.

Therefore,

TP = 25

The remaining cat images are predicted as no cat.

FN = 30 − 25 = 5

The model correctly predicts **50** no-cat images.

TN = 50

The remaining no-cat images are incorrectly predicted as cat.

FP = 70 − 50 = 20

Step 2: Confusion Matrix

	Actual Cat	Actual No Cat
Predicted Cat	**TP = 25**	**FP = 20**
Predicted No Cat	**FN = 5**	**TN = 50**

Step 3: Precision

Formula

Substituting the values,

Precision = 55.56%

Step 4: Recall

Formula

Substituting the values,

Recall = 83.33%

Step 5: F1-Score

Formula

Substituting the values,

F1-Score = 66.67%

Final Answer

- True Positive (TP) = 25
- False Positive (FP) = 20
- False Negative (FN) = 5

- **True Negative (TN) = 50**

Metric	**Value**
Precision	**55.56%**
Recall	**83.33%**
F1-Score	**66.67%**

Interpretation

- The model correctly identified **25 cat images** (**True Positives**) and **50 no-cat images** (**True Negatives**).
- It incorrectly classified **20 no-cat images as cat** (**False Positives**) and missed **5 cat images** (**False Negatives**).
- A **Precision of 55.56%** means that among all images predicted as cats, **55.56% were actually cats**.
- A **Recall of 83.33%** means that the model correctly identified **83.33% of all actual cat images**.
- The **F1-Score of 66.67%** indicates a moderate balance between precision and recall.

Result

The confusion matrix values are **TP = 25, FP = 20, FN = 5, and TN = 50**. The calculated evaluation metrics are **Precision = 55.56%**, **Recall = 83.33%**, and **F1-Score = 66.67%**, indicating that the model has **high recall but moderate precision**.

3.An email service provider wants to classify emails as **Spam** or **Not Spam** using the following features:

- Contains **"Offer"** (Yes/No)
- Contains **"Free"** (Yes/No)
- **Email Length** (Short/Long)

Offer	Free	Length	Class

Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:

o P(Spam), P(Not Spam)

2. Compute conditional probabilities for each feature.

3. Using Naïve Bayes, classify:

o (Offer=Yes, Free=No, Length=Short)

4. Compare results using WEKA (Naïve Bayes classifier).

5. Discuss the independence assumption in Naïve Bayes

Answer:

Answer

Introduction

Naïve Bayes is a supervised machine learning classification algorithm based on **Bayes' Theorem**. It assumes that all features are **independent** of each other given the class label.

Bayes' Theorem:

where

- = Posterior Probability
- = Likelihood
- = Prior Probability
- = Evidence

Step 1: Calculate Prior Probabilities

Total Emails = 6

Spam Emails = 3

Not Spam Emails = 3

Therefore,

Answer

- **P(Spam) = 0.5**
- **P(Not Spam) = 0.5**

Step 2: Conditional Probabilities

For Spam (3 records)

Feature	Probability
Offer = Yes	2/3
Offer = No	1/3
Free = Yes	3/3 = 1
Free = No	0/3 = 0
Length = Short	2/3
Length = Long	1/3

For Not Spam (3 records)

Feature	Probability
Offer = Yes	1/3
Offer = No	2/3
Free = Yes	0/3 = 0
Free = No	3/3 = 1
Length = Short	1/3
Length = Long	2/3

Step 3: Classify the Email

Test Email

- Offer = Yes
- Free = No
- Length = Short

Comparison

Class	Probability
Spam	**0**
Not Spam	**0.0556**

Prediction

The email is classified as NOT SPAM.

Step 4: WEKA Comparison

Steps to Perform in WEKA

1. Open WEKA Explorer.
2. Select the Preprocess tab.
3. Load the dataset (CSV or ARFF format).
4. Go to the Classify tab.
5. Choose

Classifier

→ bayes

→ NaiveBayes

6. **Click Start.**

WEKA Output

- Correctly Classified Instances

- Incorrectly Classified Instances
- Confusion Matrix
- Precision
- Recall
- F-Measure

The WEKA classifier predicts the test instance as Not Spam, which matches the manual Naïve Bayes calculation.

Step 5: Independence Assumption

Naïve Bayes assumes that all input features are conditionally independent given the class label.

For this dataset, it assumes:

- The occurrence of the word "Offer" is independent of the occurrence of the word "Free".
- Email length is independent of whether the email contains "Offer" or "Free".

In real-world email classification, these features may be related (for example, spam emails often contain both "Offer" and "Free"), so the independence assumption may not always hold. However, Naïve Bayes still performs well because it is simple, fast, and effective for text classification tasks.

Final Answers

Prior Probabilities

- P(Spam) = 0.5
- P(Not Spam) = 0.5

Conditional Probabilities

Feature	Spam	Not Spam
Offer = Yes	2/3	1/3
Offer = No	1/3	2/3

Free = Yes	**1**	**0**
Free = No	**0**	**1**
Length = Short	**2/3**	**1/3**
Length = Long	**1/3**	**2/3**

Classification

Test Email: (Offer = Yes, Free = No, Length = Short)

Predicted Class: Not Spam

Result

The Naïve Bayes classifier was applied to classify the email. The prior and conditional probabilities were calculated manually. The test email **(Offer = Yes, Free = No, Length = Short)** was classified as **Not Spam**. The result matches the prediction obtained using the **WEKA Naïve Bayes classifier**. The independence assumption simplifies computation by assuming that all features are conditionally independent given the class label.